

- 9 (a) e.g. reduces gain
increases bandwidth
less distortion
greater stability(1 each, max 2) B2 [2]
- (b) gain = $-R_F / R_I$
= $-8.0 / 4.0$ M1
numerical value is 2 A0 [1]
- (c) (i) 2, 6 and 7 A1 [1]
- (ii) e.g. digital-to-analogue converter (allow DAC)
adding / mixing signals with 'weighting' B1 [1]

[Total: 5]

- 10 (a) (i) e.m. radiation / photons is produced whenever a charged particle